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Project 4: The General Knowledge Quiz  
DecodeLabs Python Programming - Industrial  
Training Kit
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Goal: Ask 3 questions, keep a score counter  
(+1 per correct answer),  
and print the final score at the end.
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Key Skills:
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- If-Else logic & Variables (Control Flow)  
- Input sanitization (.strip() and  
.lower()) to defeat the  
  case-sensitivity / whitespace trap  
- The Accumulator Pattern for state  
management (score = 0, then score += 1)  
- f-strings for clean, dynamic output  
formatting
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"""
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# Each question block: (question text,  
correct answer, already sanitized to  
lowercase)
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```
QUESTIONS = [  
    ("What is the capital of France?",  
     "paris"),  
    ("Which planet is known as the Red  
Planet?", "mars"),  
    ("What is the chemical symbol for  
water?", "h2o"),
```

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]
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```
def ask_question(question, correct_answer):
    """
    Repeatable Question Block (the four-
    step micro-architecture):
    1. Ask & Capture 2. Sanitize 3.
    Evaluate 4. Execute
    Returns True if the answer is correct,
    False otherwise.
    """
    raw_answer = input(f"{question} ")

    # Sanitize: strip stray whitespace,
    then lowercase for caseless matching
    sanitized_answer =
    raw_answer.strip().lower()

    if sanitized_answer == correct_answer:
        print("Correct! ✅\n")
        return True
    else:
        print(f"Incorrect. The correct
        answer was '{correct_answer}'.\n")
        return False

def main():
    print("=== DECODELABS GENERAL KNOWLEDGE
    QUIZ ===\n")

    score = 0 # State Initialization:
    integer enforces strict data integrity
```

```
total_questions = len(QUESTIONS)

for question, correct_answer in
QUESTIONS:
    if ask_question(question,
correct_answer):
        score += 1 # Accumulator:
state only increments on success

# Output: f-string interpolation for
clean final feedback
print("--- QUIZ COMPLETE ---")
print(f"Your final score:
{score:>2}/{total_questions}")

if score == total_questions:
    print("Perfect score! 🏆")
elif score >= total_questions / 2:
    print("Good job! 👍")
else:
    print("Keep practicing! 📖")

if __name__ == "__main__":
    main()
```